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REVIEW

Machine Learning for Prediction in Psychotherapy and Psychiatry

A narrative review of treatment response, dropout and suicide risk prediction

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ABSTRACT

Machine learning promises to improve clinical prediction in mental health care by detecting complex patterns in large data sets. This narrative review examines three areas in which such predictions have been studied intensively: response to treatment and the selection of treatments for individual patients, dropout from psychotherapy, and suicide risk. Studies in all three areas report that models can discriminate better than chance, and in some cases better than conventional regression approaches. Several limitations, however, constrain the clinical value of these findings. Many models have been developed and tested within single data sets, and external validation has shown that apparent accuracy may not transfer to new trials, settings or populations. For rare outcomes such as suicide death, even models with good discrimination yield very low positive predictive values, so that most patients flagged as high risk will not experience the event. Prospective studies demonstrating that model-guided decisions improve outcomes remain scarce, and algorithms can reproduce biases embedded in their training data. We discuss reporting and appraisal standards such as TRIPOD and PROBAST, decision curve analysis, and the conditions under which prediction tools may support, rather than replace, clinical judgment and the therapeutic encounter.

Keywords: machine learning · clinical prediction models · treatment selection · suicide risk · positive predictive value

1 Introduction

Clinicians constantly make predictions: whether a patient is likely to benefit from a particular treatment, whether therapy will be discontinued prematurely, and whether a person in crisis is at imminent risk. Machine learning, understood broadly as a family of methods that learn predictive relationships from data with limited prior specification, has raised hopes that such judgments can become more accurate and more individualized (Chekroud et al., 2021). The growth of electronic health records, routine outcome monitoring and large trial data sets has made these methods increasingly feasible in psychiatry and psychotherapy.

This narrative review summarizes the state of research in three areas: treatment response and treatment selection, dropout, and suicide risk. It does not aim at an exhaustive synthesis; rather, it highlights representative studies and the methodological issues that determine whether a prediction model can be clinically useful. The guiding distinction is between statistical performance, typically expressed as discrimination and calibration, and clinical utility, that is, whether using a model leads to better decisions and outcomes than not using it (Fusar-Poli et al., 2018; Steyerberg, 2019).

2 Treatment Response and Treatment Selection

Research on treatment selection asks which of several available treatments is most promising for a given person. Cohen and DeRubeis (2018) reviewed approaches in depression that combine multiple patient characteristics into individualized recommendations and stressed that such approaches must be tested in samples independent of those used to build them. An influential early example used data from a large pharmacological trial to predict remission under citalopram and tested the model in an independent trial (Chekroud et al., 2016). Accuracy was reported as modestly above chance in the external sample for some treatment arms but not for another, a

pattern that already illustrated both the promise and the limits of the approach.

Later work has sharpened these limits. Chekroud et al. (2024) examined models predicting outcomes across several independent clinical trials of antipsychotic medication for schizophrenia. Models performed well within the trial in which they had been developed but, according to the authors, no better than chance when applied to other trials, and pooling data across trials did not resolve the problem. The authors described this as illusory generalizability. Although these findings concern pharmacological treatment, they bear directly on psychotherapy, where samples are often smaller and treatments less standardized. They also suggest that internal cross-validation, however carefully performed, cannot substitute for testing in genuinely independent samples.

The decisive test for treatment selection is prospective. Lutz et al. (2022) randomized therapist–patient dyads in an outpatient cognitive behavioral therapy clinic to a digital decision support system or to treatment as usual. They reported better outcomes when therapists followed the recommended strategy in early sessions, but no main effect of the feedback component, and they emphasized that benefits depended on the quality of implementation. Such studies show that the value of a model lies not in its accuracy alone but in how it is used.

3 Predicting Dropout

Premature termination is common in psychotherapy and is associated with poorer outcomes, which makes it an obvious target for prediction. In a large naturalistic sample of outpatients receiving cognitive behavioral therapy, Bennemann et al. (2022) compared a range of algorithms using nested cross-validation and reported that an ensemble of tree-based and nearest-neighbor methods identified dropout better than a conventional generalized linear model. At the same time, overall predictive accuracy remained modest, and the authors noted that not all algorithms were suited to naturalistic data and binary events.

Two conclusions follow. First, more complex algorithms are not automatically superior; their advantage depends on the structure of the data and the sample size, and simpler models should always be reported as a benchmark. Second, a dropout prediction is useful only if it is connected to an intervention whose effect can itself be evaluated, for example renewed attention to the therapeutic alliance or a joint review of the patient's goals. Predicting dropout without changing anything merely documents it in advance. Because dropout is defined inconsistently across studies, for instance by session counts or by therapist judgment, the outcome definition itself should be reported and justified.

4 Suicide Risk and the Problem of Base Rates

Suicide risk prediction is the area in which the gap between statistical performance and clinical utility is most visible. A meta-analysis covering five decades of research found that established risk factors predicted suicidal thoughts and behaviors only slightly better than chance (Franklin et al., 2017), which fueled hopes that machine learning applied to large data sets could do better. Models based on electronic health records have indeed reported high discrimination; in a study across several large health systems, Simon et al. (2018) found that visits with the highest risk scores accounted for a substantial share of subsequent suicide attempts and deaths.

Discrimination, however, does not determine how many flagged patients will actually experience the outcome. That quantity, the positive predictive value, depends heavily on the base rate. A hypothetical illustration makes the point: for an outcome occurring in one of 1,000 patients, a test with 90% sensitivity and 90% specificity would flag about 100 patients, of whom roughly one would experience the event. Consistent with this logic, a systematic review of suicide prediction models reported good global classification metrics but very low positive predictive values, particularly for suicide death (Belsher et al., 2019). Even in the study by Simon et al. (2018), the large majority of patients in the highest risk stratum did not attempt suicide during the follow-up period.

This does not render such models worthless. A low positive predictive value may be acceptable if the resulting action is low in cost and harm, such as a structured conversation about safety, and unacceptable if it triggers coercive measures. Decision curve analysis offers a way to weigh benefits and harms across a range of decision thresholds (Vickers & Elkin, 2006). The ethical question is therefore not only how accurate a model is, but what follows from its output for the person concerned.

5 Generalizability, Bias and Clinical Utility

Across all three areas, the most frequent weakness is insufficient validation. Models developed and evaluated within one data set tend to overestimate their performance, especially when the number of

candidate predictors is large relative to the number of events. Reporting guidelines such as TRIPOD (Collins et al., 2015) and appraisal tools such as PROBAST (Wolff et al., 2019) set out what readers need to know: how participants were selected, how predictors and outcomes were defined, how missing data were handled, and whether internal and external validation were performed. Calibration, the agreement between predicted and observed risks, deserves as much attention as discrimination, because poorly calibrated risk estimates can mislead decisions even when patients are ranked correctly (Steyerberg, 2019).

Generalizability also has a social dimension. Obermeyer et al. (2019) showed that a widely used health care algorithm, which used costs as a proxy for health needs, systematically underestimated the needs of Black patients. Analogous risks exist in mental health care, where access to treatment, diagnostic practices and documentation differ across groups. Models should therefore be evaluated for performance across relevant subgroups, and the choice of target variable should be examined critically. Finally, clinical utility requires prospective evidence, ideally from randomized comparisons of care with and without the model, as well as attention to how clinicians and patients understand and act on predictions (Chekroud et al., 2021; Fusar-Poli et al., 2018). Where predictions are shown to patients, their meaning and uncertainty should be explained in understandable terms.

6 Conclusion

Machine learning has produced a substantial body of prediction research in psychotherapy and psychiatry, and some models discriminate better than chance or better than conventional approaches. Yet evidence that their use improves outcomes remains limited, their performance often fails to generalize, and for rare events their positive predictive value is necessarily low. Progress depends less on more powerful algorithms than on careful validation, transparent reporting, attention to fairness and prospective evaluation of clinical utility. For psychotherapy research, this implies closer collaboration between clinicians, methodologists and data scientists.

From a meaning-centered perspective, a further caveat applies. Frankl insisted that a person is never fully determined by conditions but retains the freedom to take a stand toward them (Frankl, 1969). A risk score describes a statistical tendency in a population; it does not decide what an individual will make of his or her situation. Prediction tools can sharpen clinical attention, but they should remain in the service of the encounter between patient and therapist rather than take its place.

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